



AI and Entrepreneurship

Hackathon & HealthTech

Startup Week

AI & Entrepreneurship Day One

Today's Takeaways

01

AI at the Core of Modern Startups

Why AI is the engine of high-value ventures today

02

AI Propels Every Startup Stage

From problem discovery to go-to-market

03

The Startup Journey: Lean Canvas

Problem → Customer → Solution → Revenue → Investment

04

HealthTech Hackathon Challenges & Prompting that Works

Where AI is critical for each of the 7 real-world problems

A low-angle, close-up shot of a person's legs and feet while running on a track. The runner is wearing dark-colored athletic shoes with a prominent, light-colored, grid-like tread pattern on the soles. The background is heavily blurred with horizontal streaks of yellow, orange, and blue, suggesting high speed and motion. The overall lighting is warm, with a strong yellow/orange glow from the left side.

Where do you start?

WHEN YOU ARE IN A HURRY

| The AI HealthTech Startup "Win-Win-Win" Strategy

Focus on 4 Fs = Perfect PITCH/DEMO

- ✓ **Facts:** Ground your technical claims and AI architecture in rigorous scientific peer-reviewed data.
- ✓ **Frustrations:** Pinpoint the human and operational bottleneck (e.g., clinician burnout, rural access).
- ✓ **Financial Impact:** Demonstrate how your technology decreases overhead, proving business model viability.
- ✓ **Finish Line:** Clearly define the ultimate clinical and commercial objective of your technology.

MISSION FOR AI HACKATHON & HEALTHTECH STARTUP WEEK

“
*Build something meaningful
with people from different
backgrounds... and turn
research & AI into
something that can make a
real impact in the world.*”

The Human-AI Venture Framework

Entrepreneurial Stage	Role of Artificial Intelligence	Primary Human Task (Scientist-In-The-Loop)
Idea Generation	Acts as an expansive co-creator; generates adjacent applications & patent alternatives.	Critically filters clinical feasibility; verifies alignment with physical laws and lab restrictions.



**START
WITH
THE
FINISH LINE**



Business Plan & Pitch

Building before you have a TEAM plan is a waste of time.

Let CUSTOMERS be your guide.



Key Components of a Business Plan & Pitch Presentation

The following slides must be part of your presentation.

The order should be close to this because this format is highly tested and is used to successfully raise capital for startups.

You have 7 slides and topics to cover in 7 minutes, roughly.

STEP ONE: Get out the post-its and work on your business plan.

STEP TWO: Once you have a general consensus, seek customer input to test your assumptions.

STEP THREE: Build your Slides, Hone, Iterate and Practice.

LEAN CANVAS: An Iterative Team Exercise



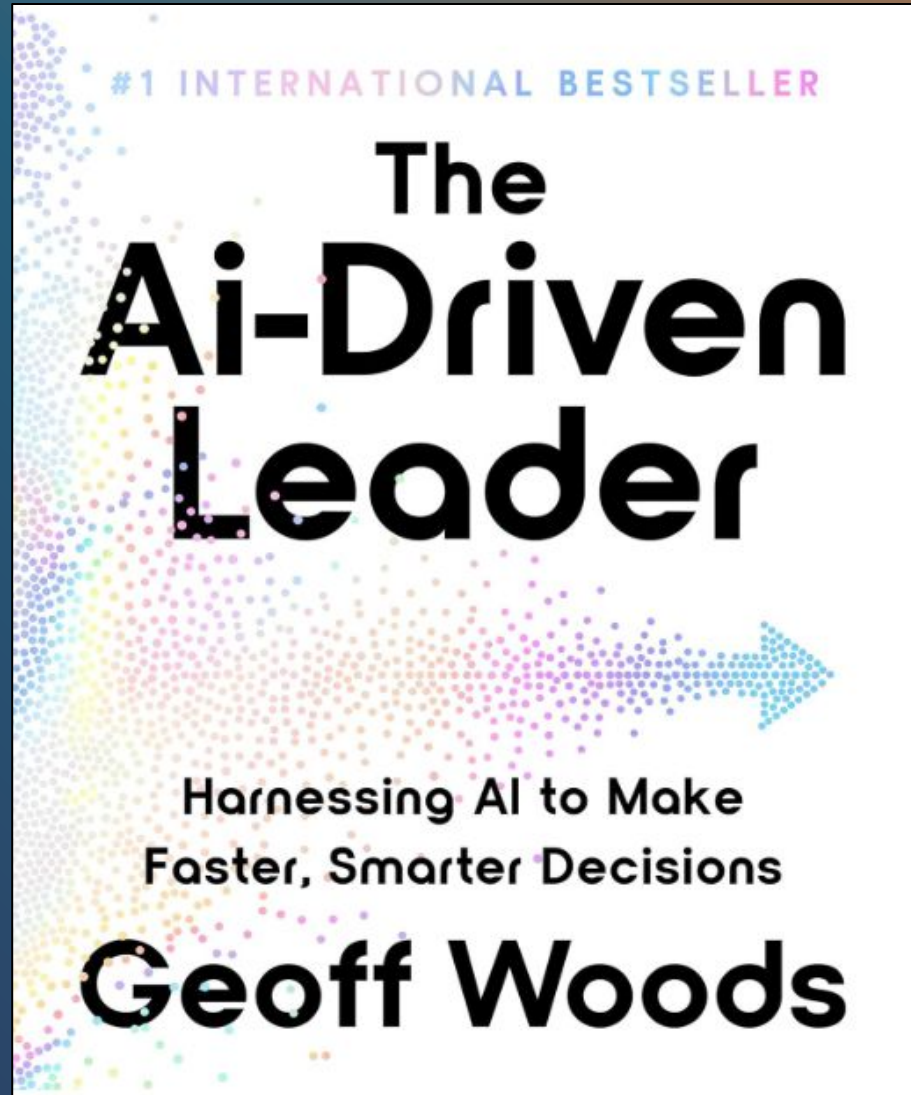
LEAN CANVAS

PROBLEM List your customers top 3 problems 2	SOLUTION Outline possible solution for each problem 4	UNIQUE VALUE PROPOSITION Single, clear, compelling that turns an unaware visitor into an interested prospect 3	UNFAIR ADVANTAGE Something that can't be easily copied or bought 9	CUSTOMER SEGMENTS List your target customers and users 1
EXISTING ALTERNATIVES List how these problems are solved today	KEY METRICS List key numbers telling how your business is doing today 8		HIGH LEVEL CONCEPT List your X for Y analogy (e.g. YouTube = Flickr for videos)	CHANNELS List your path to customers 5
COST STRUCTURE List your fixed and variable costs 7		REVENUE STREAMS List your sources of revenue 6		

prompt

AI Prompting: CRIT Framework

Grow Your Prompting Expertise



INTRODUCING CRIT

*Your 4-step framework for talking to AI — Avoid **Hallucinations** & **Corporate Fluff**.*

C

CONTEXT

Arkansas students are currently facing an unprecedented mental health crisis, characterized by a decade-long average delay between the onset of symptoms and first treatment. [upload challenge]

R

ROLE

Be a **Product Researcher/Designer** interviewing an expert or representative of these three primary customer personas:

- The Student (User)
- The Teacher (Facilitator)
- The Parent/Caregiver (Support)

I

INTENT - INTERVIEW

The goal of this interview is to uncover the specific **behavioral signals** and **incentives** required to bridge the gap between "detecting risk" and "building resilience."

T

TASK

Generate a set of discovery questions designed to validate your assumptions about these customers re:

1. **Daily Routines:** How can resilience training be integrated into a standard Arkansas classroom routine without adding to teacher burnout?
2. **Signal Detection**
3. **Gamification/Incentives**
4. **Actionability**

Train

TALKING TO HUMANS

Success starts with understanding
your customers

GIFF CONSTABLE

with Frank Rimalovski
illustrations by Tom Fishburne
and foreword by Steve Blank

THE SEQUEL TO THE AWARD-WINNING BOOK *TALKING TO HUMANS*

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PITCH SLIDE 1: Company/Brand

- Company Name
- Branding
- List all Team Members
- Tagline or Short statement of purpose of company

Challenge 05: Improving Maternal and Infant Health in Rural Arkansas

Challenge Author: Allan Nichols, CEO of Mainline Health Systems, Inc.

Coaches: Alese Johnston, Tim Reardon, Allison Thompson Meeting Room: 106 B

Arkansas has the highest maternal mortality rate in the country, more than double the national average, and the geography of care is a central reason why. As of early 2025, only 33 hospitals in the state offered labor and delivery services, and 60 percent of rural hospitals provided none at all. The March of Dimes classified more than half of Arkansas counties as maternity care deserts in 2024, meaning they lack both a birthing facility and obstetric providers. Since 2022, three additional hospitals have ended labor and delivery services.

For women in rural Arkansas, the nearest delivery hospital may be 45 minutes or more away. Long travel times, provider shortages, fragmented care coordination, and declining access to services contribute to poorer outcomes for mothers and infants.

The underlying economics compound the crisis. Current reimbursement models often fail to reward improved outcomes, preventive care, or better coordination between providers. Hospitals bear the cost of maintaining readiness, while payment systems provide little incentive to reduce NICU admissions, increase prenatal engagement, or improve long-term maternal health outcomes.

This challenge asks student teams to explore how artificial intelligence, data, care coordination, and technology-enabled healthcare delivery can improve maternal and infant health outcomes in rural Arkansas while making more effective use of existing healthcare resources.

Team 5: Challenge 05 (Improving Rural Maternal Health)

1. **Ali Rizwan** (Cabot, Male, Python Coding, Logical Problem-solving, Tech)
2. **Alyson Chapman** (Centerton, Female, Engineering, Neural Networks, Tech)
3. **Garay Gulmammadov** (Jonesboro, Male, Web & Graphic Design, Presentation, Creative)
4. **Isidora Adeola** (Little Rock, Female, Health Info Education, App UI, Creative)
5. **Ivonne Barbosa-Melgarejo** (Little Rock, Female, Health Data Analysis, Storytelling, Creative)
6. **Kshitiz Khatiwada** (Monticello, Male, Data ETL Processes, Research Analysis, Tech)
7. **Pronita Ghimire** (Monticello, Female, UI Design, Public Speaking, Creative)

PITCH SLIDE 2: Problem

- Tell a story.
- What is the human dimension of the problem and why should we care?
- What data do you have to show the size of the problem?
- Show data about whether the problem is getting better or worse without your product.

Challenge 07: Advancing Black Men's Health

Challenge Author: Camille Hart, MPH, Research Program Director, UAMS College of Public Health

Coaches: Ayo Alase, Ken Hubbell, Kam Tiwarik Meeting Room: 201 G

Black men in the United States have the lowest life expectancy of any major demographic group, with mortality rates roughly 40 percent higher than those of white men — driven in significant part by preventable conditions including hypertension, cardiovascular disease, and prostate cancer. Despite these risks, Black men are 43 percent more likely than other men to seek care in an emergency room rather than through preventive or primary care settings, reflecting longstanding barriers of access, cost, distrust, and cultural distance from traditional health systems. The challenge is not primarily one of awareness — it is one of engagement in the right settings, with the right messengers, and at the right moments.

Community-based models are demonstrating a different path. The Barbershop Talk program at UAMS has screened nearly 800 men across 61 barbershops in Arkansas, using motivational interviewing and trusted community relationships to reach men who would not otherwise engage with the health system. Research on barbershop-based health interventions nationally confirms this approach: cultural institutions like barbershops, churches, and community centers are uniquely effective platforms for health promotion among Black men precisely because the barber is already a trusted presence in their lives. The opportunity now is to extend and scale that model with technology — using AI to support screening, follow-up, referral, and ongoing wellness engagement in a way that respects privacy, reduces stigma, and meets men where they already are. This challenge asks student teams to design a community-centered digital health tool that builds on the proven barbershop model to strengthen preventive care, mental health support, and health habit formation for Black men in Arkansas and beyond.

Team 7: Challenge 07 (Black Men's Health)

1. **Amir Houston** (Little Rock, Female, NSBE Leadership, Project Management, Creative)
2. **Bryan Smith** (Little Rock, Male, Graphic Technology, Photography, Creative)
3. **David Adedeji** (Pine Bluff, Male, Rapid Prototyping, Software Architecture, Tech)
4. **Duron Huggins** (Sherwood, Male, Branding, Idea Generation, Creative)
5. **Kauri Soumanou** (Hot Springs, Male, Organization, Project Planning, Creative)
6. **Latara Williams** (Sherwood, Female, Counseling, Mental Health Research, Creative)
7. **Lincoln Chitswa** (Holly Springs, Male, React/Firebase Full-stack Dev, Tech)

PITCH SLIDE 3: Customer

- Who is the customer of your product?
- Why do they need it, to make what impact?
- How big is the market?
- What are the segments of your market?
- What is your customer value proposition and what was your customer validation process?
- In other words, do you really know details about the people who need your product or are you guessing?

Challenge 03: Helping Arkansans Choose the Right Coverage

Challenge Coach: Derek Spicer and Brad Welshans, Arkansas Blue Cross Blue Shield

Coaches: Luke Irvin, Joe Felan, Billy Spann Meeting Room: 101 D

Choosing the right health insurance plan is one of the most important financial decisions a person or family makes — yet most people do it with little guidance and significant confusion. A 2024 survey found that four out of five U.S. adults do not feel confident in their knowledge of health insurance benefits, and more than a quarter say the selection process causes them real stress. Arkansas Blue Cross Blue Shield (ABCBS) serves hundreds of thousands of individual and family members across the state, many of whom navigate their options through digital shopping tools. Today, a member visiting arkansasbluecross.com is directed to a shopping and enrollment platform where they can compare individual and family plans—but the experience of sorting through multiple policy options, coverage tiers, and cost structures remains difficult for most people without a trusted guide.

This challenge is inspired by the way platforms like Amazon use purchase history and personal preferences to surface relevant recommendations—applied here to health insurance. The goal is to design an AI-powered tool that uses information a member already has or can easily provide — such as health history, family size, hobbies/activities, occupation, prescription needs, financial situation, and personal priorities — to suggest the ABCBS plan that is the best fit for them. Think of it as a smart, conversational product recommender built for healthcare: one that simplifies a complex decision, builds member confidence, and connects people to coverage that genuinely matches their needs. The solution should be designed with accessibility in mind, serving Arkansans across a range of ages, digital literacy levels, and life circumstances.

Team 3: Challenge 03 (Health Plan Product Picker)

1. ~~Dilasa~~ **Dilasa Yaman** (North Little Rock, Female, UI/Web Design, Logo & Branding, Creative)
2. **Eric Wang** (Cave Springs, Male, Writing, Branding, Logo Design, Creative)
3. **Ishmam Solaiman** (Little Rock, Male, AI Native Development, App UI & Design, Tech)
4. **Krish Mehta** (Conway, Male, Java Coding, UI Prototyping in Figma, Tech)
5. **Pfeiffer Slaveski** (Centerton, Female, DECA, Business Development, Planning, Creative)
6. **Sri Harshitha Kancheti** ~~Kancheti~~ (Little Rock, Female, Data Analysis, QA, AI Integration, Tech)
7. **Yogi Patel** (Little Rock, Male, Aerospace Research, Real-time AI, Tech)

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PITCH SLIDE 4: Solution

- What is your solution?
- Why is it better than current alternatives or the competition?
- How does it work?
- What, generally, does it do?
- (More details of this in the demo.)

Challenge 04: Music Therapy for Older Adults

Challenge Author: Dr. Kalyani Dhusia, UAMS, Computational Biologist

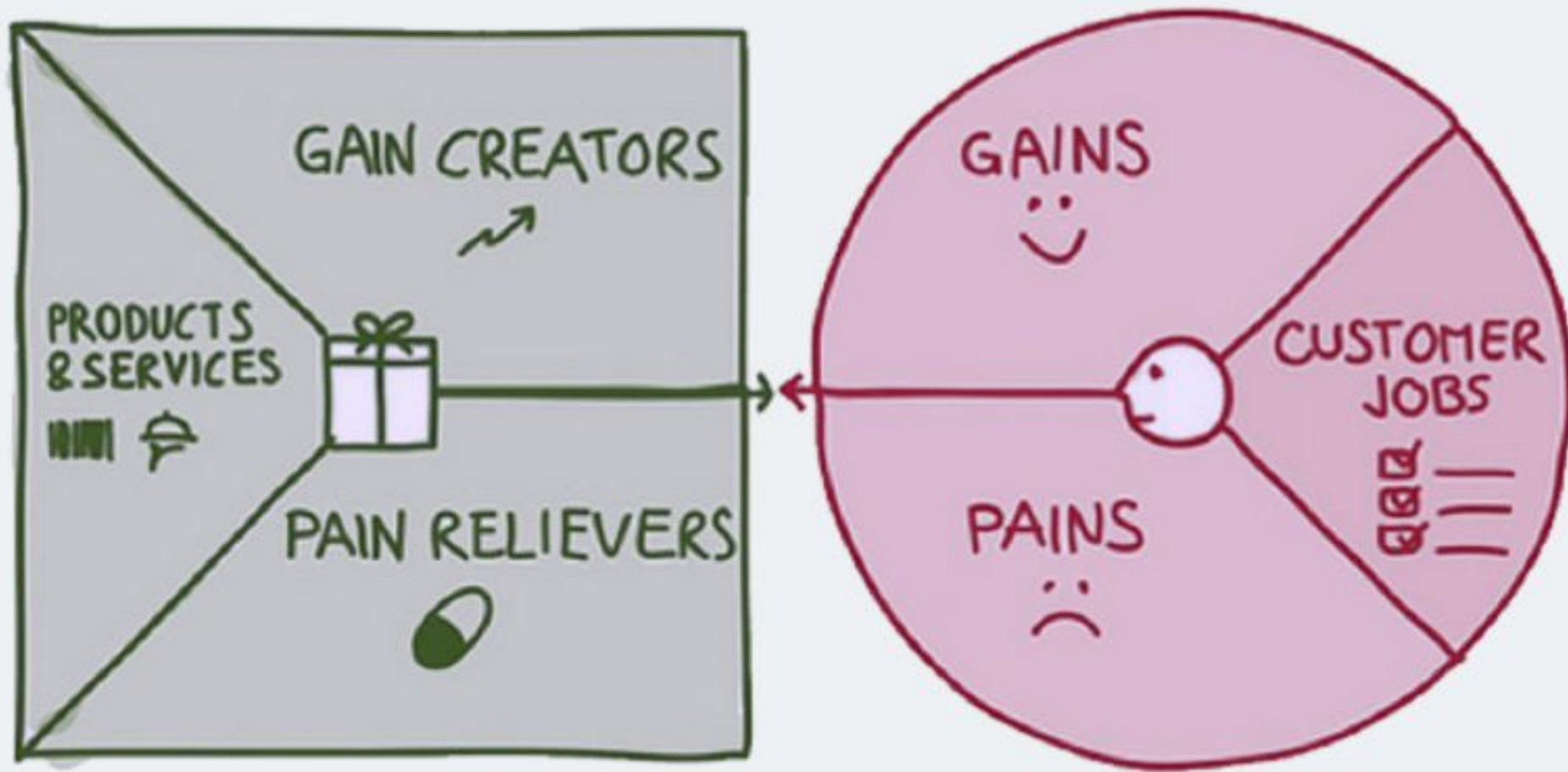
Coaches: Dr. Kalyani Dhusia, Jay Dempsey **Meeting Room:** 106 A

Music therapy is a powerful, evidence-supported approach for supporting older adults living with dementia and those experiencing social isolation. Familiar and personally meaningful music has been shown to reduce agitation, anxiety, and depressive symptoms while supporting memory and mood—even in advanced stages of cognitive decline. However, the benefits of music therapy are frequently limited to scheduled, in-person sessions led by trained professionals. For the millions of older adults, caregivers, and small care settings without regular access to a music therapist, there are few structured, easy-to-use ways to bring personalized, therapeutic music into daily life. While digital solutions could bridge this gap, most mainstream music applications are generic, are not accessible for older users, and are not built around specific therapeutic goals.

This challenge asks student teams to design and prototype an accessible digital tool that delivers personalized, music-based therapeutic engagement to older adults living with dementia or experiencing social isolation. The tool should center creative engagement and be effortlessly usable by older adults, family caregivers, or care staff—whether it functions as an intuitive playlist-builder with caregiver guidance, an interactive reminiscence experience, or a companion tracker for care facility activities. Student submissions must include a working prototype rooted in user-centered design and a viable business-and-impact case. Crucially, because this challenge serves a vulnerable population, teams must design with strict adherence to ethics, accessibility, and user dignity, ensuring the tool acts as a support system for wellbeing rather than a medical or diagnostic device.

Team 4: Challenge 04 (Music-Therapy for Dementia)

1. **Sally Alazab** (Sheridan, Female, Web Design, IT, Graphic Design, Creative)
2. **Koree Arndt** (Bella Vista, Male, Project Management, Agile Scrum, AI Music, Tech)
3. **Tanvi Virani** (Lamar, Female, Branding, Logo Design, Communication, Creative)
4. **Aras Erdag** (Little Rock, Male, Machine Learning, Music Text-classification, Tech)
5. **Cami Gonzalez** (Bentonville, Female, Pitching, Marketing visuals, social media, Creative)
6. **Joshua Davis** (Conway, Male, Logo and Web Design, Art, Creative)
7. **Satya Achanti** (Bentonville, Male, Public Speaking, Storytelling, App UI Development, Creative)
8. **Brodie Smith** (Little Rock, Male, Branding, Graphic Design, Video, Creative)



Value Proposition Map: Pains & Gains

PITCH SLIDE 5: Marketing/Distribution

- How will customers get your product?
- What resistance is there and how will you overcome it?
- What must happen for people who need your product to access it?
- Can they afford it?
- How will you reach out to more people who need your product?

Challenge 01: Next-Generation Drug Testing with Gut Organoids

Challenge Authors: Dr. Sujan Ghosh, UA Little Rock, and Dr. Rupak Pathak, UAMS

Coaches: Femi Alayesanmi, Abdo Elfikky, Fanny Milanova **Room** 101A in Library

Every drug that reaches a pharmacy shelf was tested first—and for decades, those tests relied on flat cell cultures and animal models that do not always predict how the human gut actually responds. That gap costs lives and billions of dollars: roughly nine out of ten drug candidates that pass early lab tests ultimately fail in clinical trials, often because the preclinical model simply did not reflect human biology. A new class of research tools called organoids — miniature, lab-grown tissue structures that mimic the architecture and function of real organs—is beginning to change that. The global organoid market was valued at over 800 million dollars in 2024 and is projected to nearly triple by 2030, driven by pharmaceutical companies, cancer researchers, and federal regulators actively moving away from animal testing in favor of these more predictive, human-relevant platforms.

The intestinal organoid segment is among the fastest growing, precisely because the gut's complex structure — its microscale villi, its constant mechanical movement, its role in absorbing drugs and nutrients — is so difficult to replicate with older methods. In 2025, the FDA issued new guidance encouraging the use of human-based computational and lab models in drug submissions. This challenge asks student teams to think at the intersection of bioengineering, design, and commercialization: how might a more accurate, scalable, and manufacturable gut model be brought to market in a way that genuinely advances research and reduces the cost of discovering treatments that work? Your demo should show how a pharmaceutical researcher, biotech lab, or clinician would actually use an AI-powered system to replace inefficient drug testing workflows.

Team 1: Challenge 01 (Gut-Mimetic/Organoid Testing)

1. **Sharafat Hossain** (Little Rock, Male, Computer Vision, Machine Learning, Intelligent Vision, Tech)
2. **Fatima Abbas** (Little Rock, Female, Business Case Development, Venture Strategy, Leadership, Creative)
3. Shrigun Dinesh Maroo (Little Rock, Male, Public Speaking, Storytelling, Business Case Development)
4. **Ahmet Yaman** (North Little Rock, Male, Product Design & Development, Full-stack Coding, Tech)
5. **Kshrugal Reddy Jangalapalli** (Bentonville, Male, Agentic AI, Code Development, Tech)
6. **Aditi Shashidhara** (Centerton, Female, Marketing, Graphic Design, Data Analytics, Creative)
7. **Sana Afsheen Shaik** (Bentonville, Female, Biomedical Machine Learning, Leadership, Tech)
8. **Shriya Sreejith** (Bentonville, Female, App Design & Development, Business Strategy, Tech)

PITCH SLIDE 6: Healthcare & Regulations

- How are you addressing regulatory compliance, cybersecurity and privacy issues with your product?
- How will you address the challenges in the healthcare ecosystem and the market?
- Show that you really understand what the drivers are in the market.
- What steps are needed to enter the market and grow?

Challenge 02: Early Mental Health Intervention

Challenge Authors: Dr. Allison Smith, UAMS Child Study Center and Dr. Jason Williams, Arkansas Children's Hospital, Research & Foundation

Coaches: Corey Boelkens, Janav Jawahar, Thomas Wallace

Meeting Room: 101 C

Arkansas students are navigating an unprecedented mental health crisis, compounded by acute provider shortages, rural access limits, and an average decade-long delay between symptom onset and first treatment. While traditional clinical screenings only capture a single snapshot in time, youth continuously exhibit daily, observable signals through their school attendance, sleep patterns, and digital interactions. AI and health informatics present a powerful opportunity to responsibly analyze these behavioral shifts, allowing systems to flag early warning signs and route critical insights to trusted adults and care teams long before a student reaches a crisis point.

However, identifying risk is only half the battle; mental wellness must also be proactively built. Evidence-based practices like regulated breathing, mindfulness, and stress management are proven to strengthen emotional resilience, yet students are rarely introduced to these tools until they are already in distress. This challenge tasks student teams with designing an AI-powered application or interactive ecosystem that bridges this gap. By utilizing existing frameworks like AR ConnectNow and the Trauma Resource Initiative for Schools (TRIS), your solution must securely and equitably detect early warning signs while making daily mental resilience training an engaging, habitual, and classroom-ready reality for every Arkansas student.

Team 2: Challenge 02 (Educators & Student Mental Health)

1. **Hannah Robertson** (Little Rock, Female, Web & UI Design, Coding, Creative)
2. **Rudra Pratap Mukherjee** (Little Rock, Male, Entrepreneurship, Start-up Strategy, Business)
3. **Tanisha Virani** (Lamar, Female, Marketing, Business Case Development, Business)
4. **Vishak Meenachi** (Little Rock, Male, Mental Health Triage Infrastructure, Tech)
5. **Luke Wang** (Little Rock, Male, Competitive Analysis, Robotics, Applications, Tech)
6. **Joel Johnson** (Bentonville, Male, AI Product Development, Business Strategy)
7. **Anay Pandit** (Benton, Male, Product Design & Development, AI, Tech)
8. **Jake Hu** (Little Rock, Male, Finance, Market Analysis, Web Development, Creative)

PITCH SLIDE 7: Building a Business

- Explain how you and your teammates learned about this challenge.
- What does the team think it will take to make the product financially viable/profitable or sustainable through some funding mechanism.
- What people, skills and/or agents do you need to make it happen?
- Does anyone on your team want to make this business happen?
- If so, why? If not, why not?

PROBLEM List your customers top 3 problems 2	SOLUTION Outline possible solution for each problem 4	UNIQUE VALUE PROPOSITION Single, clear, compelling that turns an unaware visitor into an interested prospect 3	UNFAIR ADVANTAGE Something that can't be easily copied or bought 9	CUSTOMER SEGMENTS List your target customers and users 1
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COST STRUCTURE List your fixed and variable costs 7			REVENUE STREAMS List your sources of revenue 6	

Challenge 06: Empowering Cancer Survivors

Challenge Author: Dr. Anne Mancino, Central Arkansas Veterans Health Administration (VA)

Coaches: Blake Stansbery, Meherun Shraboni, Thomas Wallace Meeting Room: 105 D

In 2025, new cancer diagnoses in the United States are projected to exceed 2 million for the first time — nearly 5,800 people every day. With earlier detection and improving treatments, more people are living longer after a cancer diagnosis than at any point in history: more than 18 million Americans now carry a history of the disease. But surviving cancer does not mean the journey ends. Survivors face ongoing health risks from the disease itself and from treatments — including cardiovascular effects, secondary cancers, fatigue, and lasting mental health challenges — that require years of coordinated follow-up care.

Yet the system for managing that follow-up is largely fragmented. Survivorship care plans exist and are recommended by organizations like the American Cancer Society, but they are rarely personalized, difficult to access, and not designed to keep pace with a patient's evolving goals and health status over time. For veterans, these challenges are compounded by the complexity of VA care systems, geographic barriers, and the heightened psychological burden many carry. AI is increasingly recognized as a promising tool in this space — capable of supporting personalized monitoring, proactive health coaching, connection to mental wellness resources, and two-way communication with care teams in ways that static paper plans cannot. This challenge asks student teams to design an **AI-powered survivorship companion** — a tool that helps cancer survivors, including veterans, manage their long-term health, stay connected to their care team, and pursue a full life after treatment, not just a longer one.

Team 6: Challenge 06 (Cancer Survivorship)

1. **Alesha Imran** (Centerton, Female, Marketing, Video Editing, Branding, Creative)
2. **Blessing Itodo** (Little Rock, Female, Technical Product Management, Business Analyst, Business)
3. **Disha Harishkumar** (Santa Clara, Female, App Development, Software Engineering, Tech)
4. **Shreyan Movva** (Bentonville, Male, Python Coding, HTML/CSS, User Experience, Creative)
5. **Umme Kalsoom** (Little Rock, Female, PhD Computer Science, AWS, AI Integration, Tech)
6. **Yahya Sridiajamerta** (Fayetteville, Male, Creative Visual Content, Video Marketing, Creative)

Product Demo



DEMO 1: Connect the Dots

- Refer to the four Fs during the Demo based on the pitch so it's clear that the pitch and demo are connected. Cover the
- Facts
- Frustrations
- Financial Impact
- Finish Line/Goals Accomplished

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8. **Shriya Sreejith** (Bentonville, Female, App Design & Development, Business Strategy, Tech)

DEMO 2: Tell the User Story

- Be clear who is using the solution and walk us through that person's "Day in the Life."
- How does it solve the users' problem?
- What does the "Before and After" look like?

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8. **Jake Hu** (Little Rock, Male, Finance, Market Analysis, Web Development, Creative)

DEMO 3: Solution Details

- How far did you get?
- What features were you able to explore?
- Show how your solution will work with some real-world examples.
- Describe the logic of your solution and why it makes an impact.
- What technologies did you use to make the prototype?

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This challenge is inspired by the way platforms like Amazon use purchase history and personal preferences to surface relevant recommendations—applied here to health insurance. The goal is to design an AI-powered tool that uses information a member already has or can easily provide — such as health history, family size, hobbies/activities, occupation, prescription needs, financial situation, and personal priorities — to suggest the ABCBS plan that is the best fit for them. Think of it as a smart, conversational product recommender built for healthcare: one that simplifies a complex decision, builds member confidence, and connects people to coverage that genuinely matches their needs. The solution should be designed with accessibility in mind, serving Arkansans across a range of ages, digital literacy levels, and life circumstances.

Team 3: Challenge 03 (Health Plan Product Picker)

1. ~~Dilasa~~ **Dilasa Yaman** (North Little Rock, Female, UI/Web Design, Logo & Branding, Creative)
2. **Eric Wang** (Cave Springs, Male, Writing, Branding, Logo Design, Creative)
3. **Ishmam Solaiman** (Little Rock, Male, AI Native Development, App UI & Design, Tech)
4. **Krish Mehta** (Conway, Male, Java Coding, UI Prototyping in Figma, Tech)
5. **Pfeiffer Slaveski** (Centerton, Female, DECA, Business Development, Planning, Creative)
6. **Sri Harshitha Kancheti** ~~Kancheti~~ (Little Rock, Female, Data Analysis, QA, AI Integration, Tech)
7. **Yogi Patel** (Little Rock, Male, Aerospace Research, Real-time AI, Tech)

DEMO 4: MVP to Road Map Plan

- Throughout the demo, explain what your team learned when you were building the prototype.
- Explain how the decisions made by the tech team were influenced by teammates' knowledge and skills, by people you spoke to and those who tested the prototype/demo.
- What would you need to do to make it an MVP?
- What do you think it would take to make it a commercially viable product?
- What integrations would be required? How do you know?

Challenge 04: Music Therapy for Older Adults

Challenge Author: Dr. Kalyani Dhusia, UAMS, Computational Biologist

Coaches: Dr. Kalyani Dhusia, Jay Dempsey **Meeting Room:** 106 A

Music therapy is a powerful, evidence-supported approach for supporting older adults living with dementia and those experiencing social isolation. Familiar and personally meaningful music has been shown to reduce agitation, anxiety, and depressive symptoms while supporting memory and mood—even in advanced stages of cognitive decline. However, the benefits of music therapy are frequently limited to scheduled, in-person sessions led by trained professionals. For the millions of older adults, caregivers, and small care settings without regular access to a music therapist, there are few structured, easy-to-use ways to bring personalized, therapeutic music into daily life. While digital solutions could bridge this gap, most mainstream music applications are generic, are not accessible for older users, and are not built around specific therapeutic goals.

This challenge asks student teams to design and prototype an accessible digital tool that delivers personalized, music-based therapeutic engagement to older adults living with dementia or experiencing social isolation. The tool should center creative engagement and be effortlessly usable by older adults, family caregivers, or care staff—whether it functions as an intuitive playlist-builder with caregiver guidance, an interactive reminiscence experience, or a companion tracker for care facility activities. Student submissions must include a working prototype rooted in user-centered design and a viable business-and-impact case. Crucially, because this challenge serves a vulnerable population, teams must design with strict adherence to ethics, accessibility, and user dignity, ensuring the tool acts as a support system for wellbeing rather than a medical or diagnostic device.

Team 4: Challenge 04 (Music-Therapy for Dementia)

1. **Sally Alazab** (Sheridan, Female, Web Design, IT, Graphic Design, Creative)
2. **Koree Arndt** (Bella Vista, Male, Project Management, Agile Scrum, AI Music, Tech)
3. **Tanvi Virani** (Lamar, Female, Branding, Logo Design, Communication, Creative)
4. **Aras Erdag** (Little Rock, Male, Machine Learning, Music Text-classification, Tech)
5. **Cami Gonzalez** (Bentonville, Female, Pitching, Marketing visuals, social media, Creative)
6. **Joshua Davis** (Conway, Male, Logo and Web Design, Art, Creative)
7. **Satya Achanti** (Bentonville, Male, Public Speaking, Storytelling, App UI Development, Creative)
8. **Brodie Smith** (Little Rock, Male, Branding, Graphic Design, Video, Creative)



There is a proven process for presenting a successful demo. Demos are the primary way that tech companies sell their products to customers. This is how companies make revenue, so this professional skillset is career-making. Demos, even demos of prototypes, are live. They are carefully orchestrated to show the people who would use the product how they would do so and with what outcome. A proven template follows:

- **Immediate clarity** - Audience understands the problem in under 30 seconds.
- **Fast “aha” moment (hook)** - Show value before features.
- **Narrative structure** - Great demos tell a transformation story. (Before/After)
- **Relatable use case** - Show a real human workflow, not abstractions.
- **Minimal cognitive load** - Simple UI, focused flow, no feature dumping.
- **Proof** – Tell that you know how it really works including details about customer feedback, speed, metrics, privacy, compliance, and integrations. Products don’t exist in a vacuum. They are part of a larger system. Show that you understand how your product fits into the system.
- **Memorability** – Share one idea so emphatically, the audience repeats it afterward.
- **Confidence + polish** – Practice smooth transitions, rehearsed timing, and make sure your tech environment is stable. You will do better to have a few, solid and memorable parts of the demo instead of trying to stuff in many features that don’t work well. Quality not quantity.

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Empowering Cancer Survivors

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